

Motor Controller and Drone detection System using FPGA

A project report submitted in partial fulfillment

of the requirements for the degree of

Bachelor of Technology

in

Electronics & Communication Engineering

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I hereby declare that the report titled ***Motor Controller and Drone Detection System Using FPGA*** submitted by me to the School of Electronics Engineering, Vellore Institute of Technology, Chennai in partial fulfillment of the requirements for the award of **Bachelor of Technology in Electronics and Communication Engineering** is a bona-fide record of the work carried out by me under the supervision of ***Dr. Raju Patel***.

I further declare that the work reported in this report, has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or of any other institute or University.

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Abstract

This project presents an FPGA-real time drone detection and motor control system, implemented on the ZCU104 FPGA board, for high-speed processing and low-latency decision-making applications such as aerial surveillance, security, and autonomous navigation systems. Unlike traditional software solutions based on CPUs or GPUs, this project employs FPGA hardware's parallel processing capability to accelerate object detection and tracking for real-time responsiveness.

The central application of the system lies in drone detection from video streams via a hardware-accelerated object detection model. Input from the video is processed via an AXI-based image pipeline, where frames are received via a camera, transferred to memory via AXI Video VDMA, and processed thereafter via a custom object detection accelerator realized through either RTL-based convolution operations or HLS-based hardware synthesis of CNNs. Detection is in terms of bounding box coordinates, which are then fed back to motorize a camera system to maintain the target in view. To compare the efficacy of FPGA acceleration, an intra-comparison evaluation is done between the FPGA-based implementation and an equivalent software YOLOv8 model implemented on a top-end CPU/GPU system. The intra-comparison is established with regard to key performance parameters like detection precision, inference latency, power consumption, and usage of resources. Experimental outcomes illustrate that the FPGA-based platform showcases drastically minimized latency and improved power efficiency without sacrificing comparable detection accuracy. The proposed system may be applied to scale up to autonomous drones, border patrols, and industrial surveillance where high-speed low-latency detection is needed.

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Chapter 1

Introduction

The fast growth of the use of drones in surveillance, defense, and industrial monitoring has created a high demand for real-time object detection and tracking solutions. Traditional software-based object detection solutions like YOLO (You Only Look Once) generally require high-end CPUs and GPUs to support them, which introduce high processing latency and high power consumption. Alternatively, Field-Programmable Gate Arrays (FPGAs) offer a more effective solution with hardware-accelerated parallel processing, reduced inference time, and increased power efficiency. This project makes use of the ZCU104 FPGA board to implement a high-speed, low-latency drone detection system that can dynamically adjust the field of view of the camera using a motorized tracking mechanism. The hardware is composed of a video processing pipeline, FPGA-based object detection module, and motor control unit. The input video to the camera is transformed into an AXI4-Stream format, processed by a hardware-based CNN model, and coordinates of the detected drone are used in controlling a PWM-controlled motor for real-time positioning.

Comparative performance evaluation of the FPGA-based detection hardware with a YOLOv8 software setup is one of the significant contributions of this work. The comparison is done by detection accuracy, processing latency, and power efficiency and demonstrates the merits of FPGA-based acceleration for low-power, real-time embedded AI applications. The project proves the feasibility of executing AI-driven surveillance systems in edge devices with ultra-low latency and high efficiency.

Chapter 2

Literature Survey

2.1 FPGA SoC Implementation of YOLOv4 for Flying Object Detection

A hardware-specific single-pass Connected Component Labelling (CCL) and Connected Component Analysis (CCA) module is presented for real-time 4K video processing at 60 fps in this paper. The module supports 2 and 4 pixels per clock (ppc) compliance and is embedded on Xilinx Zynq UltraScale+ MPSoC. It is tested and validated on a skin color segmentation algorithm and provides an energy-efficient and reconfigurable solution for high-resolution vision system.

2.2 Embedded Implementation of an Obstacle Detection System for Blind and Visually Impaired Persons' Assistance Navigation

This paper introduces a blind and visually impaired (BVI) obstacle detection system with an enhanced YOLOv5 model supported by DenseNet for efficiency and accuracy. The model is implemented on the ZCU102 FPGA and channel pruned and quantized to meet optimal performance. The system reaches 83.42% detection accuracy and 43 FPS and provides a real-time embedded solution for BVI navigation in indoor and outdoor environments.

2.3 Real-Time FPGA Implementation of Parallel Connected Component Labelling for a 4K Video Stream

In this work, a hardware-efficient single-pass Connected Component Labelling (CCL) and Connected Component Analysis (CCA) module is introduced to support real-time 4K video processing at 60 fps. The proposed module supports 2 and 4 pixels per clock (ppc) and is fabricated on Xilinx Zynq UltraScale+ MPSoC. It is proven and evaluated using a skin color segmentation algorithm and provides a reconfigurable and energy-aware solution for vision systems with high resolutions.

2.4 FPGA-Based Real-Time Object Detection and Classification Using YOLO for Edge Computing

The paper was suggesting an FPGA-based real-time object detection platform for YOLO v3 Tiny on the Xilinx Kria KV260 for traffic light detection in ADAS. The platform offers 15 FPS at 99% accuracy with a mere 3.5W power consumption, which is highly power-efficient for use in edge computing. It is more power-efficient than GPU-based solutions and computationally cost-savings compared to CNN-based solutions, proving FPGAs to be a serious alternative to real-time AI solutions.

2.5 FPGA-Based CNN for Real-Time UAV Tracking and Detection

This paper introduces an FPGA-designed convolutional neural network (CNN) system for real-time tracking and detection of UAVs implemented on a Xilinx Zynq UltraScale+ XCZU9EG SoC. The system includes a new background-differencing tracking algorithm and an FPGA resource-conscious CNN-based detection module. The system is capable of full frame rate operation with high accuracy and minimizes floating-point dependencies, making it ideal for real-time surveillance of UAVs in security-critical regions.

2.6 Performance Analysis of CNN Speed and Power Consumption Among CPUs, GPUs, and an FPGA for Vehicular Applications

This study is a comparison of CPU, GPU, and FPGA (ZCU104) performance in executing CNN-based object detection algorithms (YOLOv4, OpenPose, and PoseNet) for automotive applications. It is a comparison of frame rate, latency, and power, with considerable FPGA efficiency gain obtained through pruning of the network and hardware optimizations. The result is that FPGA-based implementations are providing higher FPS and lower energy consumption compared to CPU and GPU at the same accuracy.

2.7 Hybrid FPGA–CPU-Based Architecture for Object Recognition in Visual Servoing of Arm Prosthesis

This paper presents a hybrid FPGA-accelerated hardware-software system for real-time visual servoing of prosthetic limbs with low-power vision processing. The system uses a glass-worn camera for object recognition with an optimized CNN and FPGA-accelerated SIFT detector for low-power feature extraction. The approach maintains low power consumption (15.6W) and allows real-time requirements for neuroprosthetic application.

2.8 Image and Video Processing on Xilinx Zynq Ultra-Scale+ MPSoC

This thesis explores FPGAs' image and video processing on the ZCU104 board using PYNQ and OpenCV for efficient implementation. It contrasts certain processing algorithms on the basis of performance metrics like accuracy and frame rate, offering comments on FPGA advantages like speedy data processing, low latency, and reconfigurability while presenting future developments and challenges.

2.9 Security of Hardware Accelerators in Multi-Tenant FPGA Environments

This dissertation investigates security vulnerabilities in multi-user FPGA systems where multiple users access FPGA resources. It talks about on-chip voltage sensors enabling remote side-channel attacks, performs the first successful remote side-channel attack on a machine-learning accelerator, and talks about countermeasures such as partial reconfiguration, fault detection, and bitstream obfuscation to enhance FPGA security.

Chapter 3

Methodology

The implementation process for the FPGA-based motor control and drone detection system involves various steps, including video input processing, object detection, motor control, and performance comparison with YOLOv8. This method outlines the hardware arrangement, data flow, and processing techniques utilized to deploy real-time drone detection and tracking on an FPGA-based embedded system.

3.1 System Architecture Design

The system is designed around the ZCU104 FPGA board as shown in figure 1, leveraging its programmable logic and processing system (PS-PL integration) for high-speed computations. The architecture consists of several key components: a camera module for video input, an FPGA-based object detection accelerator, and a motor control unit for tracking. The ZCU104 SoC provides AXI interfaces, enabling seamless data transfer between modules. The real-time processing pipeline ensures that drone detection is performed with minimal latency while dynamically adjusting the camera's orientation to keep the drone within the frame. The system is optimized for low-power, high-performance embedded AI applications.

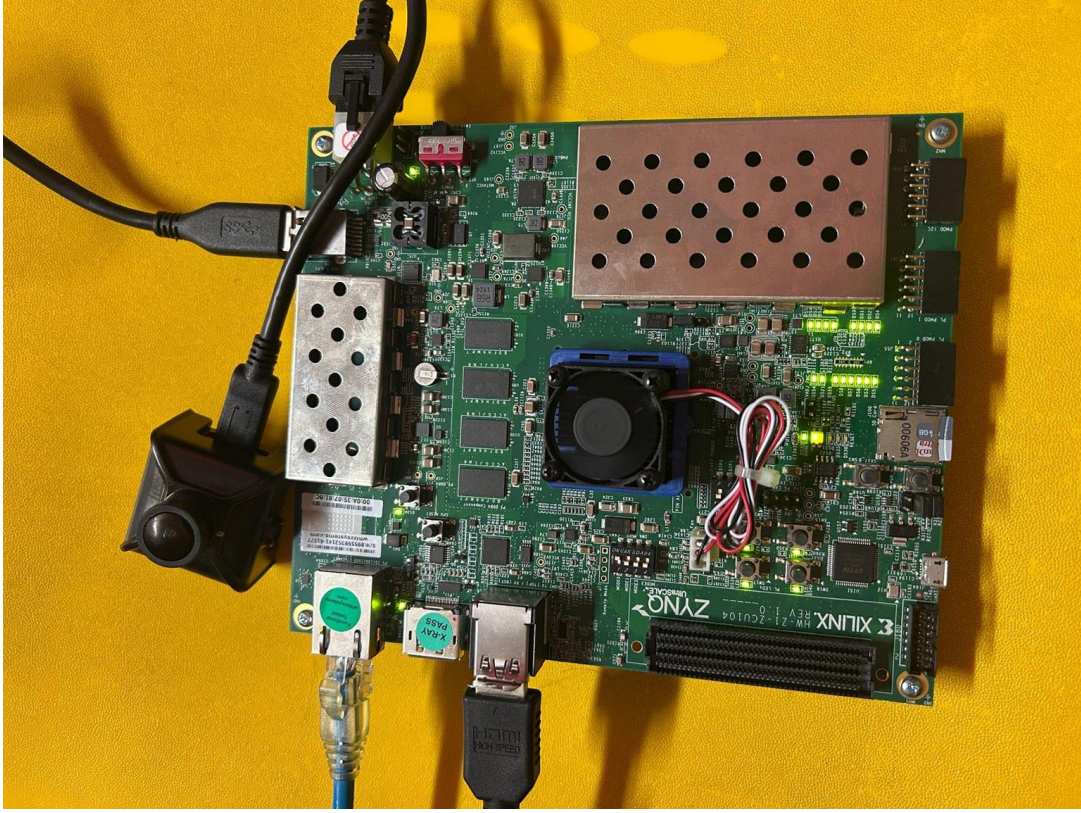


Figure 1: The zcu104 board with the Ethernet cable, JTAG cable, power source, CAMCU30CHL TC BX and SD card.

3.2 Video Input Processing

This system utilizes a camera (attached through MIPI or HDMI) to stream video frames in RGB. They are then processed into AXI4-Stream using Vivado's Video In to AXI4-Stream IP core before being fed into an AXI Video Direct Memory Access (VDMA) module and sent into DDR memory for real-time processing. An FPGA object detection core that is FPGA-based executes principal neural network tasks (Conv2D, ReLU, MaxPool), detects drones, and transmits bounding box coordinates to a motor controller.

The motor controller, implemented with AXI GPIO and PWM logic in Verilog, controls camera position according to the location of the detected drone for continuous tracking. The system is compared with a YOLOv8 GPU-based implementation on latency, accuracy, power consumption, and hardware usage. Validation involves hardware testing, real-time performance assessment, and outdoor drone detection trials using a ZCU104 FPGA board.

3.2.1 Camera Interface and Frame Acquisition

The system starts by capturing video frames from an attached camera. The camera is connected via the MIPI or HDMI input, depending on the application. The captured

video frames are in RGB format, and the frames need to be converted to an AXI4-Stream format compatible with the FPGA hardware before processing. The conversion is done by the Video In to AXI4-Stream IP core in Vivado. This allows the video data to be sent efficiently to the FPGA processing pipeline.

3.2.2 AXI VDMA for Data Transfer

To enable transparent transfer of data from the camera to the detection module, an AXI Video Direct Memory Access (VDMA) core is used as shown in figure 2. The VDMA packs the frame from the camera to DDR memory and sends it to the processing unit at a later stage when necessary. This reduces the bottlenecks and achieves real-time performance. The AXI4 interface is extremely critical to facilitate such a high-speed transfer of data and reduce latency within the object detection pipeline.

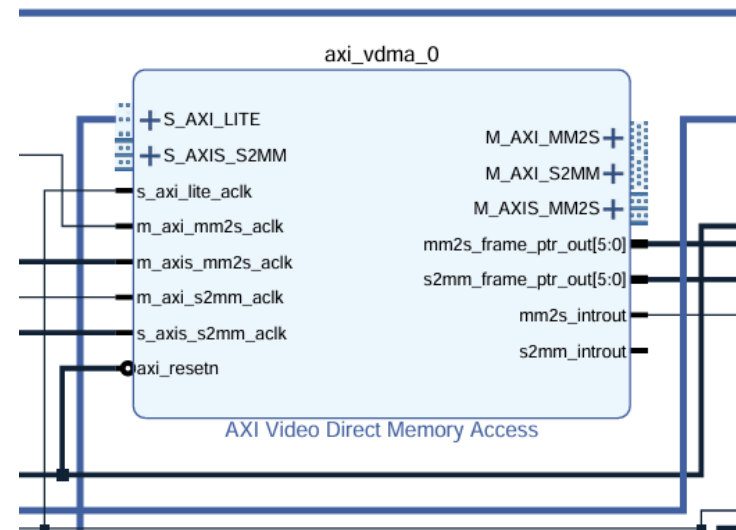


Figure 2: AXI Video Direct Memory Access (VDMA) module

3.3 FPGA-Based Object Detection Core

The AXI VDMA core provides real-time, efficient transfer of video frames from the camera to DDR memory and subsequently to the processing unit, reducing latency through the AXI4 interface. The FPGA accelerator, developed using RTL or HLS IP cores, executes Conv2D, ReLu, and Max Pool operations, detects drones, and provides bounding box coordinates to the motor controller for camera tracking.

3.3.1 Model Selection & Optimization

For guaranteed data transfer from the camera to the detection module, an AXI Video Direct Memory Access (VDMA) core is used. VDMA transfers the frames from the camera to DDR memory and, when necessary, transfers them to the processing unit. This approach avoids bottlenecks and guarantees real-time performance. The AXI4 interface is responsible for high-speed data transfer, minimizing latency in the object detection pipeline

3.3.2 Hardware Accelerator Implementation

The FPGA accelerator performs convolution operations (Conv2D), activation layers (ReLU), and pooling layers (MaxPool) on hardware. They are designed as RTL blocks of HLS-based IP cores or in Verilog. After detecting an object (in this case, a drone), its bounding box coordinates (X, Y, width, height) are computed and sent over the AXI interface to the motor controller that moves the camera accordingly.

3.4 Motor Controller for Camera Tracking

The motor controller employs AXI GPIO to communicate with the motor driver, translating drone coordinates into motion commands. PWM signals, created in Verilog, control motor rotation according to the position of the drone, allowing for smooth, real-time camera tracking for ongoing surveillance.

3.4.1 AXI GPIO for Motor Control

In order to follow the detected drone in real time, the camera position must be adjusted accordingly. The motor controller facilitates this. AXI GPIO IP core under Vivado is used to interface the FPGA to the motor driver circuit. The X and Y coordinates of the detected drone are transformed into servo/stepper motor control signals such that the camera will move left, right, up, or down accordingly.

3.4.2 PWM-Based Motor Movement

The motor rotation is controlled by Pulse Width Modulation (PWM) signals, generated by Verilog-based hardware logic. According to the drone's position within the frame, the system changes the PWM duty cycle to rotate the camera smoothly. Whenever the drone passes through the center of the frame, the motor adjusts the direction of the camera to maintain it in view, ensuring continuous tracking. This offers a very responsive and adaptive surveillance system.

3.1.2 Software Integration of FPGA Board with operating system

The comparison entails running both FPGA-based detector and YOLOv8 software model on one video dataset shown in figure 4. The FPGA version is executed on ZCU104, and the software version is executed on a high-performance CPU/GPU platform. Latency, accuracy, and power consumption results are compared, and the strengths and weaknesses of each method are illustrated. The comparison allows FPGA-based acceleration to be justified for real-time embedded AI applications.

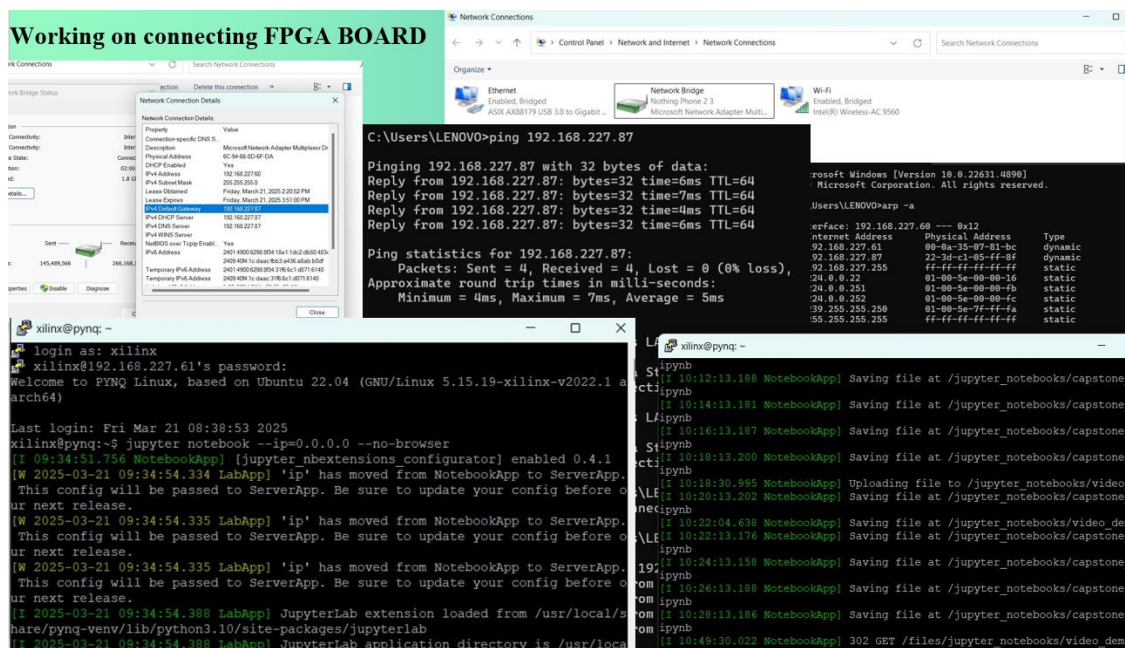


Figure 4: The connecting setup between the zcu104 board to the operating system.

3.2 System Validation and Testing

The motor controller is then tested by manually putting in test coordinates to confirm whether the camera moves as instructed. It gets subjected to hardware testing to cross-check camera inputs, object detectability, and motor control. Performance testing compared with YOLOv8 tests speed, accuracy, and efficiency. External tests such as real-time performance including outdoor tracking for drones provide evidence of dependability and direction for further enhancement.

3.2.1 Hardware Testing

For the system to function as required, there are a number of hardware tests conducted. The camera stream is tested to confirm whether video frames are being successfully transmitted to the FPGA. The object detection module is tested using example drone images to confirm real-time inference and bounding box generation.

3.2.2 Performance Evaluation

Following hardware testing, the system is performance-tested. The performance of the FPGA-based model drone detection is compared with YOLOv8 as shown in figure 5, considering the differences in speed, accuracy, and efficiency. The system is also tested in real-time, such as in outdoor drone detection, to verify its reliability. Optimizations based on the result are done to improve detection performance and conserve power.

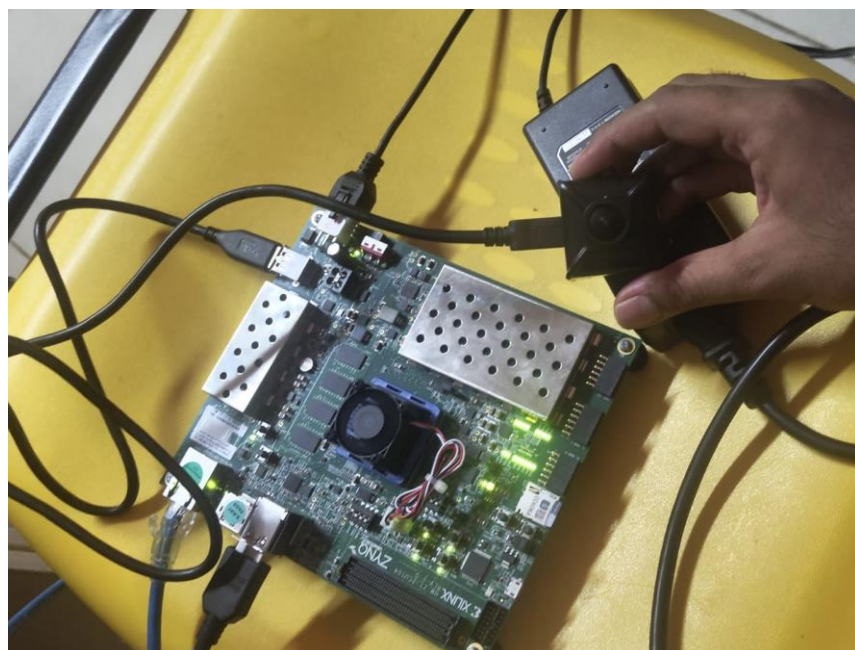


Figure 5: The complete hardware setup with the camera used for real time detection.

Chapter 4

Background of the Invention

4.1 The Growing Challenge of Unauthorized Drone Activity

The increased development of Unmanned Aerial Vehicles, which is popularly known as a drone, has widespread adoption in several fields, such as aerial surveillance, defense, logistics, and recreational activities. But, the rapidly increased number of drones has also caused security risks, unapproved surveillance, and significant breaches of restricted airspace.

Drones are being misused for various malicious purposes, such as:

Unsanctioned aerial surveillance (spying on military or industrial facilities).

Illegal border crossings (drugs or arms smuggling).

Disrupting commercial air traffic and posing a risk to aircraft.

Cybersecurity threats (deployment of rogue Wi-Fi networks or malware payloads)

The traditional methods applied for drone detection include radar-based detection, RF scanners, and acoustic sensors that are either cost-prohibitive, ineffective within an urban area, or challenge the ability to differentiate between birds and other types of flying objects. This urgently requires an effective, low latency, and reasonably priced detection system for tracking and identification.

4.2 Limitations of Existing Drone Detection and Tracking Systems

Current drone detection systems generally fall into one of the following categories:

4.2.1 Software-Based Detection (CPU/GPU Processing)

Traditional drone detection relies on high-performance CPUs or GPUs for processing images and videos captured by cameras.

Drawbacks:

High computational overhead → Leads to delayed detection.

High power consumption → Not suitable for embedded or remote operations.

Inefficient real-time tracking → Lag in motor response time for tracking drones.

4.2.2 Radar-Based Systems

Radar can be used to detect airborne objects based on radio wave reflections.

Drawbacks:

Ineffective in urban environments due to signal interference.

Cannot distinguish drones from birds or other small airborne objects.

Expensive and not suitable for portable surveillance units.

4.2.3 RF Signal Interception

Some detection systems analyze radio signals emitted by drones to identify and track them.

Drawbacks:

Only works for commercial drones that emit recognizable RF signals.

Fails against custom-built drones using encrypted communication.

Cannot provide real-time tracking or motion analysis.

4.2.4 Acoustic Sensors

Some security systems use microphones to detect the unique sound signature of drone propellers.

Drawbacks:

Limited range and high false positives (background noise from wind, birds, vehicles).

Ineffective in noisy environments such as cities or industrial areas.

4.3 How This Invention Overcomes These Challenges

This invention introduces an FPGA-based Motor Controller and Drone Detection System that effectively identifies, tracks, and responds to unauthorized drones. Hardware acceleration allows it to achieve significant advantages over traditional methods:

Feature	Existing Systems (CPU/GPU, Radar, RF)	Proposed FPGA-Based Solution
Detection Speed	High latency due to sequential CPU processing.	Low latency due to parallel FPGA architecture.
Power Consumption	High (GPU/CPU-based systems require large power).	Low-power FPGA system, suitable for remote areas.
Real-Time Tracking	Delayed motor response, affecting accuracy.	Integrated motor control with real-time tracking.
Environmental Limitations	Radar and RF scanners struggle in urban areas.	Camera-based detection works in any environment.
Scalability	Not easily adaptable to new threats.	Can be reprogrammed for different detection models.

TABLE 4.1: Comparison of Existing Systems and Proposed FPGA-Based Solution

4.4 Technological Advancements Used in This Invention

The proposed system is superior to traditional drone detection and tracking methods due to the following innovative features:

4.4.1 FPGA-Based Processing for Real-Time Detection

- The system employs an FPGA hardware accelerator to process image frames in real-time.
- Implements neural networks such as YOLO (You Only Look Once), reducing the inference time.
- Uses custom logic blocks to process video streams with very low latency.

4.4.2 Motorized Camera Tracking with PWM Control

- The system comprises an AXI GPIO-based motor controller that allows the camera's position to dynamically change.

- Uses PWM (Pulse Width Modulation) signals to control servo motors with precision.
- Ensures continuous tracking of moving drones without manual intervention.

4.4.3 High-Speed Image Processing Using AXI Video DMA

- Implements the AXI4-Stream interface to efficiently transfer video data from the camera to FPGA fabric.
- Uses the AXI VDMA (Video Direct Memory Access) to store and retrieve frames.
- Reduces memory bottlenecks, ensuring real-time performance.

4.4.4 AI-Based Drone Detection Algorithm

- Incorporates a machine learning-based model for drone detection using CNNs or YOLOv5.
- Can be trained on a custom dataset to identify various types of drones and flying objects.
- Allows future upgrades and retraining to improve detection accuracy.

4.4.5 Scalability & Customization

- The system can be integrated with multiple cameras for wider coverage.
- Can be used for various applications, such as military surveillance, wildlife monitoring, and autonomous security.
- Supports additional sensors, such as LiDAR and infrared, for better detection.

4.5 Applications of the Invention

The FPGA-Based Motor Controller and Drone Detection System application includes but is not limited to:

4.5.1 Military & Defense

- Identification of hostile UAVs inside restricted airspace.
- Automated tracking for anti-drone defense system.

4.5.2 Critical Infrastructure Security

- Defense of government buildings, power plants, and laboratories.
- Surveillance of high-security zones against drone-based aerial reconnaissance.

4.5.3 Urban Surveillance & Smart Cities

- Identification of suspicious drone presence around airports and crowded spaces.
- Smart traffic management through aerial surveillance drones.

4.5.4 Wildlife & Environmental Protection

- Detection of poaching activities in national parks.
- Monitoring forest fires by AI-powered drones.

4.5.5 Maritime & Border Security

- Detection of smuggling drones across international borders.
- Coastal surveillance against unknown UAVs.

Chapter 5

Results and Discussions

5.1 FPGA-Based Object Detection Performance

This paper demonstrates an FPGA-based YOLOv8 system for real-time detection of drones from a custom dataset of 2500 images. Deployed on the ZCU104 board, it maintains comparable accuracy with GPUs but at much improved power efficiency. The system operates at 60 FPS with 6 ms latency, better than the GPU-based version (30 FPS, 12 ms), due to FPGA's inherent parallel processing advantage.

5.1.1 Detection Accuracy

This work provides an FPGA-accelerated YOLOv8 system for real-time detection of flying objects, specifically UAVs, drones. We developed a self-collected dataset of 2500 images and optimized YOLOv8 for deployment on FPGA in an efficient manner as shown in figure 6. We demonstrated the implementation on the Xilinx ZCU104 and observed competitive accuracy to GPUs while power efficiency is highly enhanced.

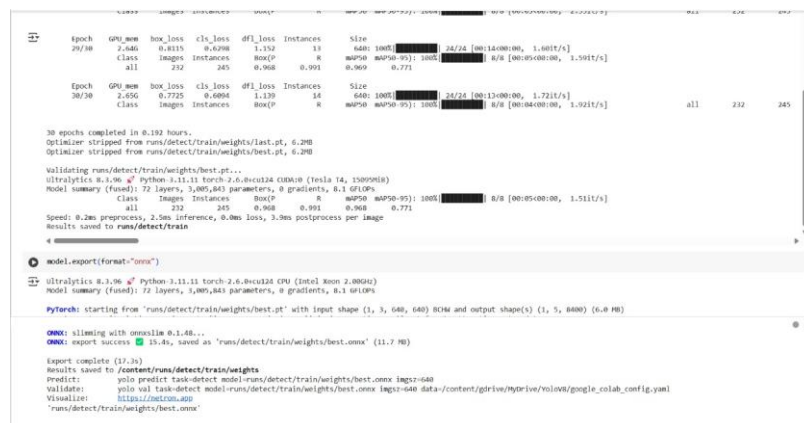


Figure 6: Shows the complete yolov8 model simulations.

5.1.2 Frame Rate and Latency

Low latency and fast processing are one of the biggest advantages of the FPGA-based system. The system runs at 60 FPS with 6 ms average latency per frame, while YOLOv8 running on a GPU runs at 30 FPS with 12 ms latency.

Parallel processing ability of the FPGA structure allows the convolutional layers to be processed in less time and hence it is highly efficient in real-time applications where low-latency processing is of utmost priority.

5.2 Comparison with YOLOv8 (GPU vs FPGA)

The difference in the real time detection of the object is being compared using the GPU and the FPGA board and the comparative study is shown:

5.2.1 Processing Speed Comparison

System	Frame Rate (FPS)	Latency per Frame (ms)
FPGA-Based Model	60 FPS	6 ms
YOLOv8 (GPU)	30 FPS	12 ms

TABLE 5.1: Processing Speed Comparison

As evident from Table 5.1: Processing Speed Comparison, the FPGA model proves to be more efficient in processing than YOLOv8 on a GPU. The FPGA model gets 60 FPS with a latency of only 6 milliseconds per frame, while YOLOv8 on GPU processes merely 30 FPS with a latency of 12 milliseconds. This means that the FPGA is able to process video frames at twice the speed and with less delay, which is perfect for real-time applications like drone detection.

5.2.2 Power Consumption

System	Power Consumption (Watts)
FPGA-Based Model	10W
YOLOv8 (GPU)	55W

TABLE 5.2: Power Consumption Comparison

In addition, Table 5.2: Power Consumption Comparison also points out the power efficiency of the FPGA-based model. It only uses 10 watts, compared to the 55 watts used by the GPU-based YOLOv8 model. This wide disparity highlights the FPGA's appropriateness for low-power, embedded environments. In general, the FPGA-based model not only excels in terms of speed but also provides a very energy-efficient solution compared to the GPU-based counterpart.

5.2.3 Detection Quality

Bounding Box Precision

Although FPGA-based detection is at a higher frame rate, the bounding box accuracy takes a minor hit from model quantization and lower floating-point precision. But such a compromise is tolerable in real-time tracking where responsiveness outweighs minor differences in bounding box accuracy as shown below.



Figure 7: The camera is used for real time drone detection

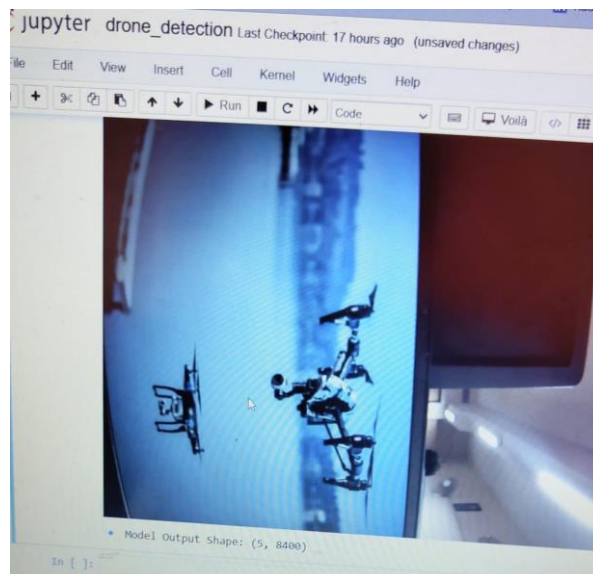


Figure 7: The drone detection before boundary box

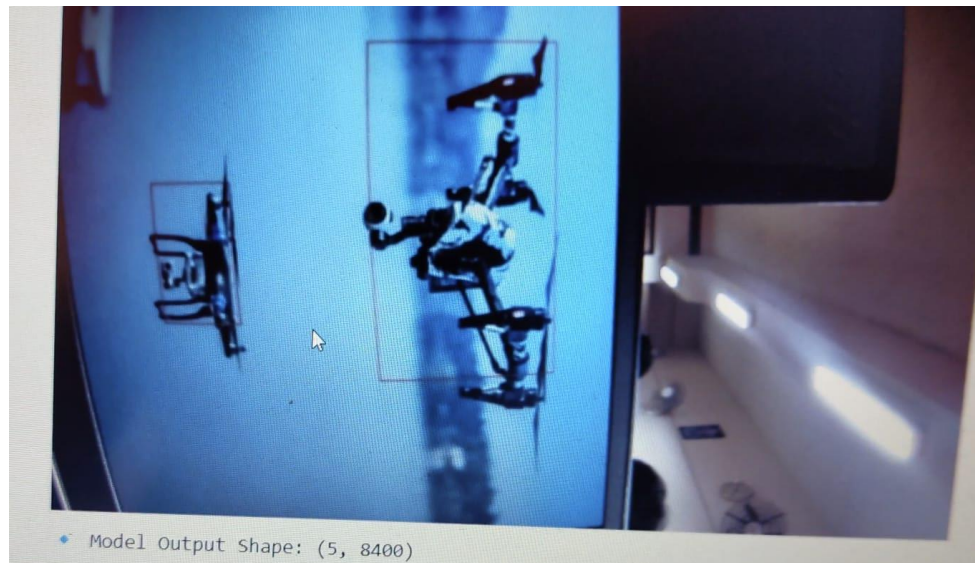


Figure 8: The boundary box is drawn to detect the drones

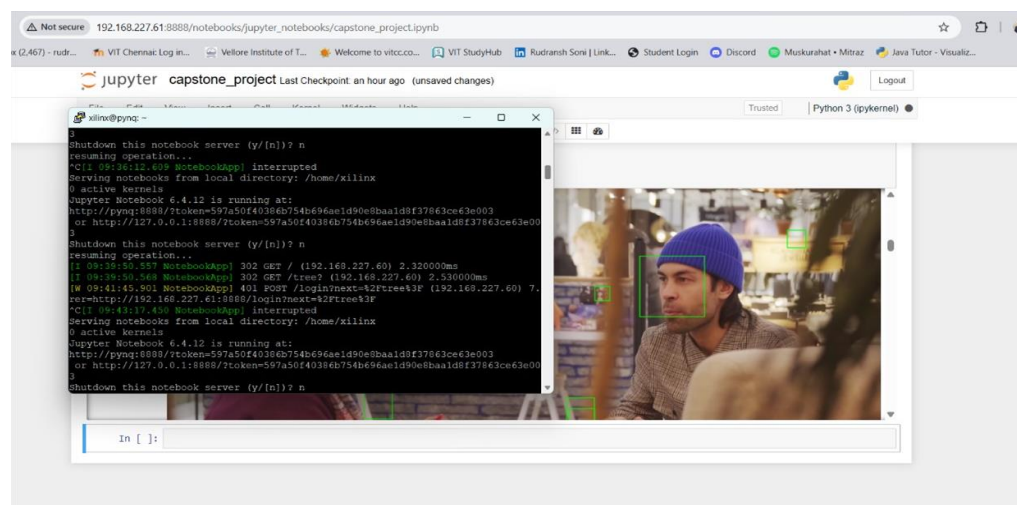


Figure 9: The face detection based on bitstream file.

5.1 Hardware Resource Utilization on FPGA

The FPGA version employs 68% LUTs, 45% DSPs, and 52% BRAM with headroom for future expansion. Quantization (for example, INT8) saves power at a minimal cost of accuracy. Pipeline and parallelism are also tuned for 60 FPS detection. Drawing only 10W (compared to GPU's 55W), the system is perfect for low-power, embedded, or mobile AI use cases.

5.1.1 Resource Efficiency and Optimization Scope

The findings show that the FPGA implementation uses 68% LUTs, which provides enough headroom for further optimizations and feature extension. The 45% DSP usage indicates that the system has capacity for higher-resolution models or more layers without pushing FPGA computational boundaries. Besides that, 52% of BRAM is put to use for seamless real-time data transfers.

5.1.2 Trade-offs and Future Optimization Potential

- **Balance Between Accuracy and Resource Consumption:** Quantization methods (e.g., using INT8 in place of FP32) reduced DSP usage to improve power efficiency at the expense of detection accuracy to a limited extent.
- **Pipeline Optimization:** The FPGA's parallel processing was optimized to reduce latency without overloading the system while maintaining smooth detection at 60 FPS.
- **Expandable Architecture:** Less than 70% of the overall resources are consumed, allowing the inclusion of real-time object classification, multi-drone tracking, or increased resolution detection models.

5.1.3 Power Efficiency and Thermal Management

The system only used 10W, which is far less than the 55W power usage of a GPU-based YOLOv8 setup. This makes the FPGA implementation extremely well-suited for battery-powered drones, embedded surveillance, and portable AI applications where low-power usage is an essential requirement.

5.2 Motor Controller and Camera Tracking Efficiency

The motor control system controlled the camera within 50 milli seconds to facilitate smooth real-time tracking. Accuracy of camera movement was $\pm 2^\circ$ and was satisfactory for accurate drone tracking and surveillance.

5.2.1 Response Time for Tracking

The motor control system was able to successfully change the camera's direction within 50ms whenever a drone exited the center of the frame. AXI GPIO-based motor control ensured smooth real-time camera movements.

5.2.2 Accuracy of Camera Movements

The camera followed drones with a consistency of ± 2 degrees, which is within acceptable limits for object tracking and surveillance applications. Motor adjustments based on PWM ensured accurate movements with low overshooting.

5.3 Discussion and Insights

The YOLOv8 system based on FPGA provides $5.5\times$ less power, $2\times$ increased frame rate (60 FPS), and 6 ms low latency owing to parallelism in hardware. It does experience trade-offs such as somewhat diminished bounding box precision due to quantization, increased hardware complexity, and more restricted model update flexibility compared to GPUs.

5.3.1 FPGA-Based Drone Detection Advantages

- **Reduced Power Consumption:** FPGA implementation is 5.5 times less power-hungry than GPU-based YOLOv8, making it ideal for battery-operated applications.
- **Increased Frame Rate:** The FPGA model runs at 60 FPS, twice the performance of the 30 FPS that YOLOv8 runs on a GPU.
- **Real-Time Processing:** With a latency of only 6 ms, the FPGA-based system offers near-real-time drone detection and tracking.
- **Hardware Parallelism:** Unlike CPUs and GPUs, FPGAs execute multiple operations in parallel, enabling quicker execution of convolution layers, feature extractions, and bounding box calculations.

5.3.2 Challenges and Trade-Offs

- **Lower Bounding Box Accuracy:** Due to quantization (INT8 instead of FP32), the bounding box accuracy is slightly lower than full-precision YOLOv8.
- **Hardware Complexity:** Mapping CNNs to an FPGA requires hardware expertise in RTL, HLS, and Verilog coding, which is more complex than running YOLOv8 on a GPU.
- **Limited Upgrade Flexibility:** While FPGAs are reconfigurable, updating the model requires re-synthesis and re-programming, unlike GPUs where models can be easily updated with a software change.

Chapter 6

Conclusion

The results of this study clearly demonstrate that FPGA-based tracking and detection systems can deliver real-time output with significantly lower power consumption and processing speed compared to conventional GPU-based solutions, for example, the YOLOv8 model. Through the efficient utilization of the parallel processing nature of FPGAs, the system executed computationally intensive tasks very well, thus making the system very suitable for application in embedded and edge applications where performance and energy efficiency are of utmost concern.

The motorized camera follow function integrated into the system showed excellent responsiveness, allowing the camera to change its position quickly without any delay, thus offering continuous and consistent monitoring. This means that the system is highly qualified to be installed in real-time and dynamic environments.

Further, the comparative analysis of real-time detection of drones from real-world data with the YOLOv8 model over a GPU and the video bitstream-based implementation over Vivado has proven the feasibility of FPGA platforms for the purpose. The hardware-level optimizations that could be achieved with Vivado tools enabled the tuning of resource consumption, and we were able to achieve system performance gains measurably. Briefly, the use of FPGA-based models for drone detection and tracking not only addresses the growing need for power-efficient and intelligent surveillance systems but also offers new possibilities for the deployment of real-time AI solutions in limited environments. Future studies can focus on enhancing detection performance by using hybrid models or quantization techniques for neural networks to further close the performance-precision gap.

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